

File permissions in Linux

Project description

In this project, I used Linux commands to examine and modify file and directory permissions for a research team's project files. My goal was to ensure users had the appropriate level of authorization by reviewing existing permissions, removing unauthorized access, updating permissions on a hidden file, and securing a sensitive directory.

Check file and directory details

Command Used:

`ls -la`

Explanation:

The `ls -la` command lists all files and directories, including hidden files, while displaying detailed information such as permissions, ownership, file size, and modification dates.

```
researcher2@65563b353539:~$ cd /home/researcher2/projects
researcher2@65563b353539:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 15:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 16:25 ..
-rw--w--- 1 researcher2 research_team  46 Jun 28 15:35 .project_x.txt
drwx--x-- 2 researcher2 research_team 4096 Jun 28 15:35 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 28 15:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_t.txt
researcher2@65563b353539:~/projects$
```

Describe the permissions string

`-rw-rw-r--`

Explanation

Linux uses a 10-character permission string to describe the type of file and the permissions assigned to the owner, group, and others.

- The **first character** indicates the file type.
 - `-` = regular file
 - `d` = directory

The remaining nine characters are divided into three groups:

- **Characters 2–4 (`rw-`)** represent the owner's permissions.
 - `r` = read
 - `w` = write
 - `-` = no execute permission
- **Characters 5–7 (`rw-`)** represent the group's permissions.
- **Characters 8–10 (`r--`)** represent the permissions for others.

This permission string means the owner can read and write the file, members of the group can read and write the file, and everyone else can only read it.

Change file permissions

The organization does not allow **others** to have write permission on any files.

The file that violates this policy is:

project_k.txt

Current permissions:

`-rw-rw-rw-`

Others currently have write permission, which is not authorized.

Command Used

`chmod o-w project_k.txt`

Explanation

The `chmod` command changes file permissions. The option `o-w` removes the write permission from **others** while leaving all other permissions unchanged.

Updated Permissions

`-rw-rw-r--`

```
researcher2@65563b353539:~/projects$ chmod o-w project_k.txt
researcher2@65563b353539:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 15:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 16:25 ..
-rw--w---- 1 researcher2 research_team  46 Jun 28 15:35 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 28 15:35 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_t.txt
researcher2@65563b353539:~/projects$ chmod u=rw,g=r,o= .project_x.txt
```

Change file permissions on a hidden file

The hidden file `.project_x.txt` is archived. The owner should have read and write permission, the group should have read permission only, and others should have no permissions.

Current permissions:

```
-rw--w----
```

Command Used

```
chmod u=rw,g=r,o= .project_x.txt
```

Explanation

This command assigns:

- User (owner): read and write (`rw`)
- Group: read only (`r`)
- Others: no permissions

The file remains hidden because its filename begins with a period (`.`).

Updated Permissions

```
-rw-r-----
```

```
researcher2@65563b353539:~/projects$ chmod u=rw,g=r,o= .project_x.txt
researcher2@65563b353539:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 15:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 16:25 ..
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 28 15:35 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_t.txt
researcher2@65563b353539:~/projects$
```

Change directory permissions

The **drafts** directory should only be accessible by **researcher2**, the owner.

Current permissions:

drwx--x---

Currently, members of the group have execute permission, allowing them to enter the directory. This permission should be removed.

Command Used

chmod g-x drafts

Explanation

The **chmod** command removes execute permission from the group using **g-x**. The owner keeps read, write, and execute permissions, while group and others have no permissions.

Updated Permissions

Drwx-----

```
researcher2@65563b353539:~/projects$ chmod g-x drafts
researcher2@65563b353539:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 15:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 28 16:25 ..
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 28 15:35 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 28 15:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 28 15:35 project_t.txt
researcher2@65563b353539:~/projects$
```

Summary

In this project, I used Linux commands to inspect and modify file and directory permissions based on an organization's authorization requirements. I used `ls -la` to review permissions, interpreted Linux permission strings, removed unauthorized write access from a file, corrected permissions on a hidden file, and secured a sensitive directory using `chmod`. These changes demonstrate how Linux permissions help enforce the principle of least privilege and improve overall system security.

Command Order

```
cd /home/researcher2/projects
ls -la
chmod o-w project_k.txt
ls -la
chmod u=rw,g=r,o= .project_x.txt
ls -la
chmod g-x drafts
ls -la
```